

Equations for NE533

$$\begin{aligned}
Q &= E_f \times N_f \times \sigma_f \times \phi \\
\rho c_p \frac{\partial T}{\partial t} &= \nabla \cdot (k \nabla T) + Q \\
LHR &= \pi R^2 Q_{av} \\
T(r) &= \frac{Q(R^2 - r^2)}{4k} + T_s \\
\\
T_0 - T_s &= \frac{Q}{4k} R_f^2 = \frac{LHR}{4\pi k} \\
T_s - T_{IC} &= \frac{Q}{2h_{gap}} R_f = \frac{LHR}{2\pi R_f h_{gap}} \\
T_{IC} - T_{OC} &= \frac{Q}{2h_{clad}} R_f = \frac{LHR}{2\pi R_f h_{clad}} \\
T_{OC} - T_{cool} &= \frac{Q}{2h_{cool}} R_f = \frac{LHR}{2\pi R_f h_{cool}} \\
\\
h_{gap/clad} &= \frac{k_{gap/clad}}{t_{gap/clad}} \\
\\
k_{gap}(He) &= 16 \times 10^{-6} \times T^{0.79} \\
k_{gap}(Xe) &= 0.7 \times 10^{-6} \times T^{0.79} \\
k_{gap}(y) &= (1 - y) \times k_{He} + y \times k_{Xe} \\
\\
LHR \left(\frac{z}{Z_0} \right) &= LHR^0 \cos \left[\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1 \right) \right] \\
T_{cool}(z) - T_{cool}^{in} &= \frac{2\gamma}{\pi} \frac{Z_0 \times LHR^0}{\dot{m} C_p} \times \\
&\quad \left\{ \sin \frac{\pi}{2\gamma} + \sin \left[\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1 \right) \right] \right\} \\
\\
k_{ox} &= \frac{1}{A + BT} \text{ (W/cmK)} \\
\frac{1}{B} \ln \left(\frac{A + BT_0}{A + BT_s} \right) &= \frac{LHR}{4\pi} \\
A &= 3.8 + 200 \times FIMA \text{ (cmK/W)} \\
B &= 0.0217 \text{ (cm/W)} \\
\\
\sigma_\theta &= \frac{pR}{\delta} \\
\sigma_z &= \frac{pR}{2\delta} \\
\sigma_r &= -\frac{p}{2} \\
\\
\sigma_r(r) &= -p \frac{(R_o/r)^2 - 1}{(R_o/R_i)^2 - 1} \\
\sigma_\theta(r) &= p \frac{(R_o/r)^2 + 1}{(R_o/R_i)^2 - 1} \\
\sigma_z(r) &= p \frac{1}{(R_o/R_i)^2 - 1} \\
\\
\sigma &= -E \Delta T \alpha \\
\\
\sigma_r(r) &= \frac{1}{2} \Delta T \frac{\alpha E}{1 - \nu} \left(\frac{r}{R_i} - 1 \right) \left(1 - \frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right) \\
\sigma_\theta(r) &= \frac{1}{2} \Delta T \frac{\alpha E}{1 - \nu} \left(1 - 2 \frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right) \\
\sigma_z(r) &= \frac{1}{2} \Delta T \frac{\alpha E}{1 - \nu} \left(1 - 2 \frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right) \\
\\
\sigma_r(\eta) &= -\sigma^*(1 - \eta^2) \\
\sigma_\theta(\eta) &= -\sigma^*(1 - 3\eta^2) \\
\sigma_z(\eta) &= -2\sigma^*(1 - 2\eta^2) \\
\\
\sigma^* &= \frac{\alpha E (T_0 - T_s)}{4(1 - \nu)} \\
\\
\Delta \delta_{gap} &= \bar{R}_c \alpha_c (\bar{T}_c - T_{ref}) - R_f \alpha_f (\bar{T}_f - T_{ref}) \\
\\
D(t)^2 - D_0^2 &= 2M_{GB} \gamma_{GB} t \\
\\
M_{GB}(T) &= M_0 e^{(-\frac{Q}{k_B T})} \\
\\
\frac{dD}{dt} &= k \left(\frac{1}{D} - \frac{1}{D_m} \right) \\
\\
\epsilon_D &= \Delta \rho_0 \left(e^{\frac{\beta \ln 0.01}{C_D \beta_D}} - 1 \right) \\
\\
\beta &= \frac{\dot{F} t}{N_U}
\end{aligned}$$